WM360 Anatomy & Physiology III · Week 1

BLOOD

Dr. Paymon Sadrolsadot ND PhD (TCM) · VUIM · Quiz 1 — Thursday 6:35pm

This guide is weighted, not comprehensive. Across three hours Dr. Sadrolsadot repeatedly told the class which material to hold onto and which to let go. That sorting is the most valuable thing in the lecture, so it drives the layout here:

★ HIGH YIELD	He explicitly flagged it — *"important to remember," "subject to licensing exam question," "probably the most important factor."* Know these cold.
⚠ TRAP	A misconception he corrected in class, usually after a student asked. True/false questions are built out of exactly these.
In his words	His analogy or framing. If a question is worded oddly, it will be worded his way.
○ Reference only	He said don't memorise it. Greyed out on purpose — do not spend tonight here.

Format, from his own mouth: starts 6:35, mostly multiple choice, some true/false, some matching. Closed book. 5% of your grade — one of six.

1 · What Blood Actually Is

Why it counts as connective tissue

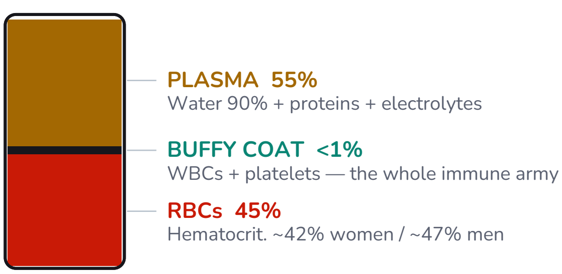
This looks like a trivia question and is really a definition question. Connective tissue is defined by **a matrix with cells suspended in it** — not by being solid. Blood has a matrix (plasma, non-living) with formed elements floating in it (living). It fits the definition exactly. Bone and cartilage feel like the intuitive answer for connective tissue; blood is the one that gets tested *because* it's counterintuitive.

> ★ **HIGH YIELD — Lymph is also connective tissue.** He said "blood and also lymph" — a matching question could pair either one.

Physical characteristics

Property	Value	Why it's asked
pH	7.35 – 7.45	Below = acidic, above = alkaline. Blood is slightly **alkaline**.
Viscosity	5× water	Comes from RBCs — see the tube below.
Volume	M 5–6 L · F 4–5 L	Varies with sex and age.
Colour	Scarlet O_2-rich / dark O_2-poor	Dark blood in skin vessels → **cyanosis**.

↑ RBC → ↑ viscosity → sluggish flow → clot risk **↓ RBC (anemia) → ↓ viscosity → poor O_2 delivery**



Spun blood. RBCs are 45% of the volume — that is why RBC count, not plasma, drives viscosity.

> ★ **HIGH YIELD** — The tube *is* the argument. RBCs occupy 45% of blood volume; every WBC and platelet in your body fits in the <1% buffy coat. That's why RBC count — and nothing else — sets viscosity.

Normal RBC count	Value	And the consequence
Women	4.2 – 5.4 million/µL	↑ RBC# → ↑ viscosity → flows **slower** → clot risk
Men	4.7 – 6.1 million/µL	↓ RBC# → ↓ viscosity → flows **faster** → poor O_2 delivery

> ⚠ **TRAP — Thick blood does not cause atherosclerosis.** A student asked this directly; he said atherosclerosis is **plaque buildup** (lipid). Viscosity affects how easily blood *moves*; plaque is a separate mechanism. Don't let a question merge them.

The 8 functions — 3 · 3 · 2

He grouped them, so learn the grouping and the members fall out of it.

Distribute (3)	Regulate (3)	Protect (2)
O_2 from lungs → cells	Body temperature	Prevent blood loss
Wastes → lungs & kidneys (CO_2 → lung; BUN/creatinine → kidney)	pH 7.35–7.45 (proteins buffer; HCO_3^- = alkaline reserve)	(platelets + plasma proteins → clot)
Hormones: gland → target organ	Fluid volume in circulation	Prevent infection (WBCs + immune chemicals)

Temperature, mechanically: superficial vessels **dilate** → more warm blood reaches the surface → heat radiates out → you cool down and look flushed. They **constrict** → blood stays deep → heat is retained. Summer skin vs winter skin.

2 · Plasma

Plasma is 55% of blood volume and **90% of plasma is water**. Everything else — nutrients, electrolytes, gases, hormones, wastes, and all the plasma proteins — shares the remaining 10%.

Plasma proteins

Albumin is the answer to most plasma-protein questions. It is ~60% of plasma protein, it is made in the **liver**, and it is the one he asked the class to name.

Protein	Share	Job	Made by
Albumin	60%	**Osmotic pressure** — holds water inside the vessel	Liver
Globulins (,)	—	Transport lipids, metal ions, fat-soluble vitamins	Liver
Globulins ()	—	Antibodies	Plasma cells
Fibrinogen	—	Becomes fibrin → the clot mesh	Liver
Clotting factors	—	The coagulation cascade	Liver

> ★ **HIGH YIELD — Plasma proteins are NOT used by cells as fuel.** He lingered here. RBCs and WBCs sit directly in a protein-rich fluid and never touch it — they use glucose, fatty acids, O_2. This is a ready-made true/false question.

Osmotic pressure → why Kwashiorkor has a big belly

Follow the chain — the exam may give you any link and ask for the next:

↓ dietary protein → ↓ albumin → ↓ osmotic pressure → water leaves the circulation → water enters the flesh → edema.

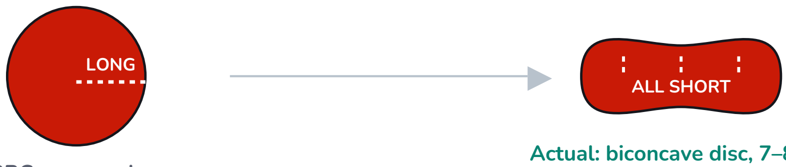
> **In his words** — On Kwashiorkor: "the big belly is not fat; it's just water — you can even see the ribs." Brittle hair too. Treatment is simply protein.

> ○ *Reference only (he said don't memorize): The remaining plasma slides — he said "these couple slides are for your reference, there are more information, you don't need to worry about that."*

3 · Red Blood Cells

Every structural feature exists to serve one job

RBCs are a case study in specialisation. Don't memorise the four features as a list — derive them from the single purpose: **carry the most oxygen possible, and deliver all of it**.



If RBCs were spheres
Core is far from surface → slow exchange

Actual: biconcave disc, 7–8 μm
Every point sits near the membrane → fast exchange

Why the dimple matters: in a sphere the core sits far from the surface; in a biconcave disc nothing does.

Feature	The trade	The payoff
Biconcave disc	Gives up volume	No interior point sits far from the membrane → fast gas exchange
No nucleus	Can't divide or self-repair; 100–120 day lifespan	97% of the cell is now hemoglobin — "bags full of hemoglobin"
No mitochondria	Must make ATP anaerobically	Two wins: more room for Hb, *and* it never burns the O_2 it carries
Flexible	—	Twists through capillaries narrower than itself

> ★ **HIGH YIELD — The RBC carries oxygen but never uses it.** He called this the elegant part of the design — the cell responsible for all the oxygen doesn't touch it. Lacking mitochondria is *why*. Expect this as a "which is true" item.

Hemoglobin

Heme = the iron-bearing part; **globin** = the protein chains. At the centre of each heme sits **one Fe^{2+}**, and that iron is what binds oxygen. This single fact is why iron deficiency makes you tired: no iron → poor heme → poor O_2 carriage → fatigue.

1 heme 1 Fe^{2+} binds 1 O_2 × **1 Hb** 4 heme → 4 O_2 × **1 RBC** 250,000,000 Hb × **= 1 BILLION O_2** carried by one cell

The number he wants you to be able to reconstruct, not just recall.

> **In his words** — "Every one red blood cell can carry 1 billion molecules of oxygen. That's how I say **we are all billionaires.**"

○ Reference only (he said don't memorize): Hemoglobin reference ranges by sex — "the number of the hemoglobin for your reference only... any time you do a test, the test will provide you the normal reference ranges."

Hematopoiesis — where blood is made

hemato = blood, **poiesis** = production. Happens **only in red bone marrow**. Yellow marrow is fat storage — and red converts to yellow as you age, which is why children produce blood more readily than the elderly.

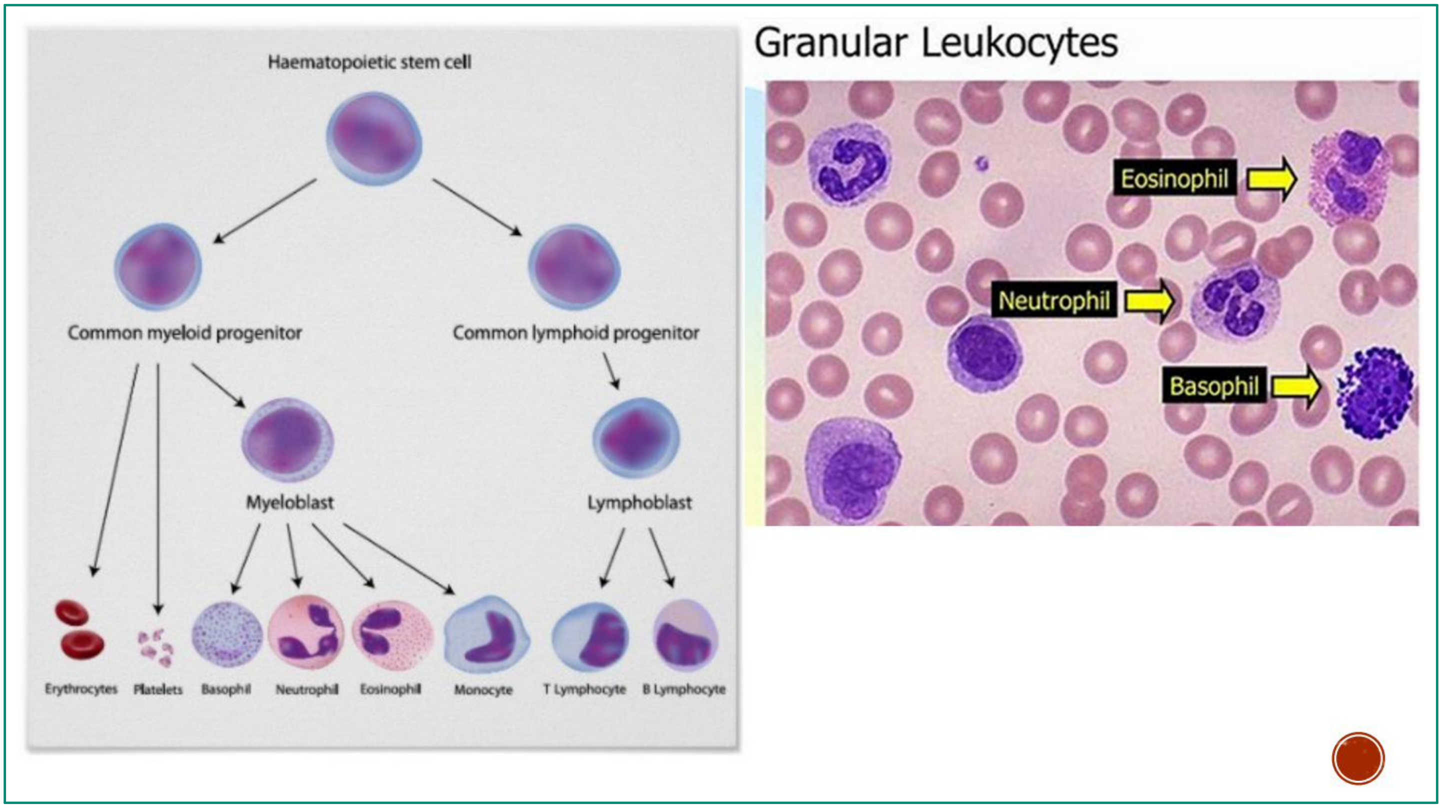
- **Sites:** axial skeleton (vertebrae), pelvic girdle, and the **proximal** ends of the long bones — proximal humerus, proximal femur.
- **Output:** ~100 billion new cells per day ≈ **one ounce (~30 mL)** of blood.
- **Stem cell:** the multipotential hematopoietic stem cell (hemocytoblast) branches into a **myeloid** line (RBCs, platelets, most WBCs) and a **lymphoid** line (lymphocytes, NK cells).

Bone marrow biopsy — how you check the factory

If the marrow makes an ounce of blood a day, the obvious clinical question is whether it's still doing that. A **large-bore aspiration needle** is driven into the **hip bone** and a sample of red marrow is drawn off to examine under the microscope.

What it answers: are the cells here normal, and is the marrow producing a healthy amount? The classic finding is **leukemia** — one cell line wildly out of proportion. His example: white cells normally sit under 10,000; a count of **70,000** tells you something is very wrong.

> ★ **HIGH YIELD — "Once committed to a pathway, they cannot change."** A myeloid progenitor can never become a lymphocyte. He said this almost verbatim — it's phrased like a true/false item.



Look for: the single split into **myeloid** vs **lymphoid progenitor**. Trace where lymphocytes sit versus where neutrophils sit — they never cross.

The fork is the whole point: one stem cell, two committed lines, and no traffic between them. Everything on the left half — RBCs, platelets, and every granulocyte — is myeloid.

> ○ *Reference only (he said don't memorize): Every intermediate name on the hematopoiesis tree — proerythroblast, promyelocyte, metamyelocyte, band cells. "At this level, this is for your information only." Recognise the shape of the tree; don't learn the rungs.*

Erythropoiesis and the reticulocyte

The one intermediate worth knowing is the **reticulocyte**: the nucleus has been ejected but ribosomes remain. It is released into blood and matures in ~2 days. Normal = **1–2%** of RBCs.

HYPOXIA — ↓ blood O_2 → **KIDNEY** — senses it, releases → **ERYTHROPOIETIN** — the hormone (EPO) → **RED MARROW** — receives the signal → **↑ RBCs** — ↑ O_2-carrying ability → **HOMEOSTASIS** — stimulus removed → *loop closes*

Negative feedback: the loop corrects itself, then the stimulus switches off.

> ★ **HIGH YIELD — EPO comes from the KIDNEYS** (liver to a small extent), and **hypoxia** is the trigger. Kidney → EPO → red marrow → more RBCs → O_2 restored → stimulus off.

> ⚠ TRAP — **A high reticulocyte count is not a disease.** A student asked; he was firm: it is a **compensatory mechanism**. The marrow is responding correctly to a problem elsewhere — bleeding, hemolysis, or iron finally arriving. The pathology is somewhere else.

> ⚠ TRAP — **Thyroid does not directly drive erythropoiesis.** Another student asked. Thyroid sets *cellular metabolic rate*; any link to RBC production is **indirect**.

What the diet has to supply

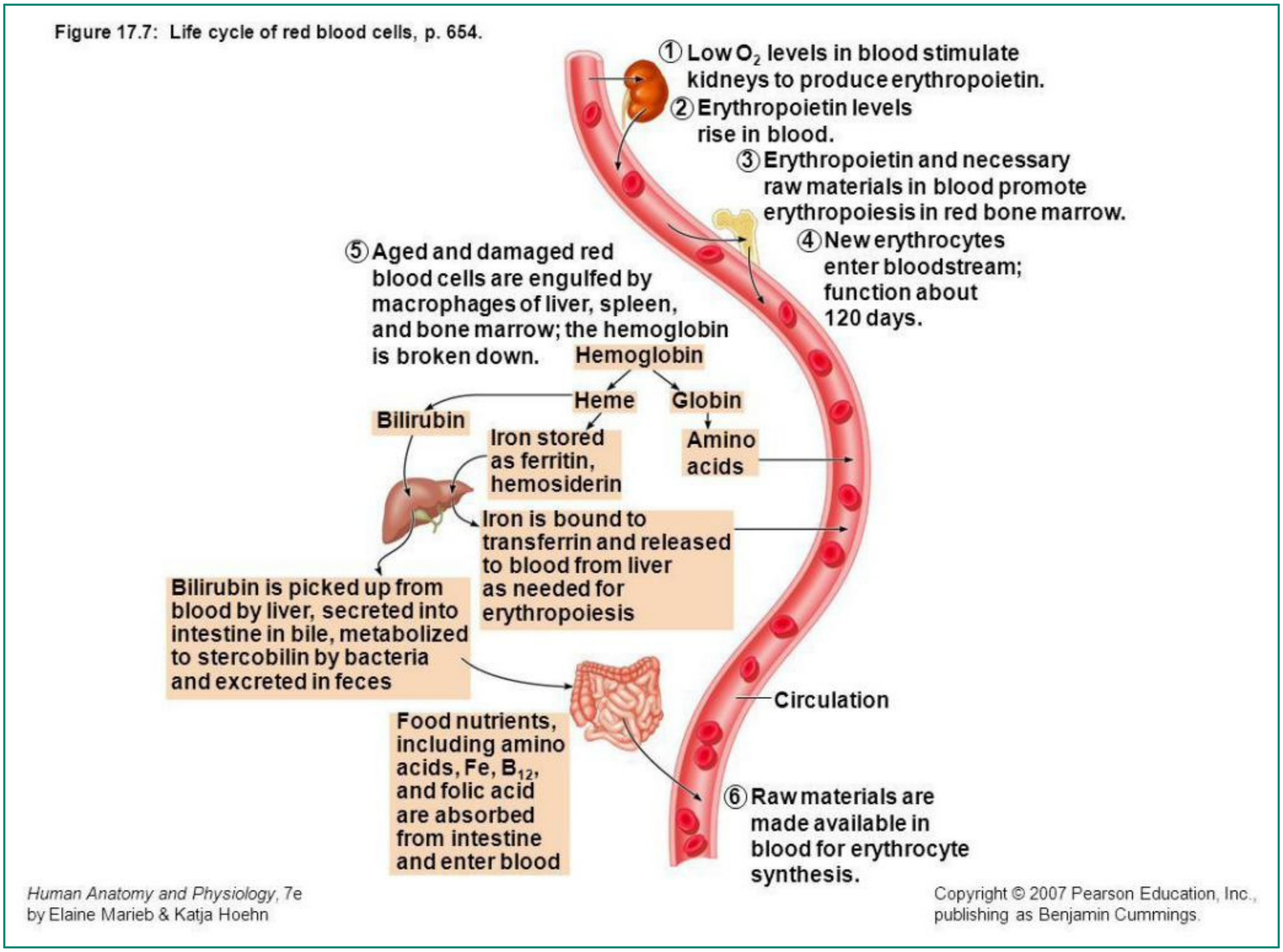
Nutrient	Because...	Deficiency
Iron	Sits at the centre of every heme; binds the O_2	Iron deficiency anemia — the most common
Amino acids	Globin is protein	Poor Hb synthesis

Nutrient	Because…	Deficiency
Lipids	Every cell membrane is a phospholipid bilayer	Poor membrane integrity
Carbohydrate	EPO itself is a glycoprotein	—
B12 / Folate	DNA synthesis for division	Pernicious / megaloblastic anemia

> **In his words** — On vegetarian and vegan diets: iron is more available from animal sources, so plant-based eaters face a **harder** time — not an impossible one. Supplementation is a valid route. He was careful not to overstate this.

The whole circuit, start to finish

He put this figure up and said: *"we see everything we discussed in one picture — the fate of production and the destruction of the red blood cells."* If you can narrate this loop out loud, §3 is done.



Look for: the six numbered steps. **1–4** is the EPO loop you already know. **5** is the spleen/liver breaking Hb into **heme + globin**. **6** is the diet feeding back in.

Production and destruction are one continuous circuit. Note that the diet inputs (bottom) and the EPO trigger (top) are the same facts from the tables above — just drawn as a loop.

> ○ *Reference only (he said don't memorize): The bilirubin tail — ferritin, hemosiderin, transferrin, stercobilin, and the route out via feces. "This is for your reference. Now you don't need to memorize this." Know that heme becomes bilirubin and leaves via the liver and gut; ignore the intermediate names.*

When RBCs go wrong

Anemia = insufficient oxygen-carrying capacity. It is a **sign**, never the diagnosis — always ask what's behind it. Three families of cause:

Cause family	Examples	Note
Blood loss	Trauma, cancer, **heavy menstruation**	★ he said this one "you need to remember"
Production fails	Iron deficiency · Pernicious (B12) · **Aplastic**	Aplastic = marrow stops making *all* lines

Cause family	Examples	Note
Destruction / defect	Sickle cell · Thalassemia	Made in normal numbers, then fail or deform

○ *Reference only (he said don't memorize): The detailed mechanism of each anemia — "we have some explanation on pathophys one, we have just kind of heard about this." Know the families and which example goes where; that's the matching question.*

Polycythemia — the mirror image

Type	Mechanism	Reversible?
Polycythemia vera	Bone marrow **cancer** — production runs out of control	No — pathological
Secondary	High altitude: thin air → hypoxia → EPO → more RBCs	**Yes** — resolves at sea level
Blood doping	Injected EPO	Detected by urine metabolites

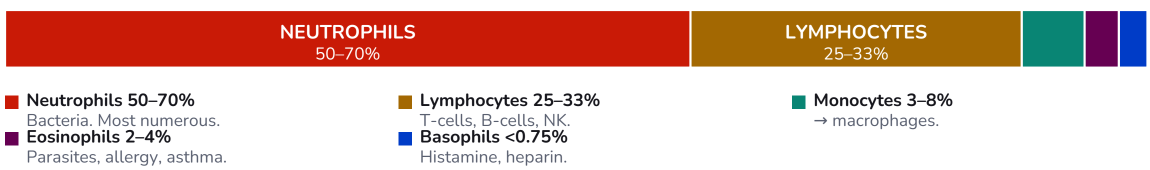
Why athletes dope, as a chain: more RBCs → more O_2 → more ATP → more energy. He walked this out explicitly. Secondary polycythemia is the same chain occurring naturally in mountain dwellers — which is why it's a **compensatory mechanism**, not a disease.

Whatever the cause, the risk is the same: ↑ RBCs → ↑ viscosity → sluggish flow → clots.

4 · White Blood Cells

★ **HIGH YIELD — WBCs are the only TRUE cells among the formed elements** — nucleus plus every organelle. RBCs have no nucleus; platelets aren't even whole cells. He stressed this phrasing, and "true cell" is precise exam language.

NEVER LET MONKEYS EAT BANANAS — rank order, most → least abundant



He said not to memorise exact percentages — but DO know the rank order and the eosinophil rule.

★ **HIGH YIELD — Eosinophilia — he named this as licensing-exam material.** Eosinophils above the normal 2–4% points to **parasitic infection, allergies, or asthma**. Think eczema, allergic rhinitis, seasonal allergy — and note it may only be raised *during* a reaction, not permanently.

Mobilisation — how a WBC reaches an infection

Term	Meaning	Why it matters
Diapedesis	"Leaping across" — squeezing through the vessel wall into tissue	The infection is *outside* the vessel. RBCs cannot do this; they stay in circulation.
Chemotaxis	Following a chemical trail to the microorganisms	Gets them to the right spot once out

Counts that move

- **Normal:** roughly 4,000–11,000/µL — a genuinely wide range.
- **Leukocytosis** = raised. Can **double within one hour** on exposure.
- **Leukopenia** = lowered. Classically **drug-induced** — a student supplied the answer he wanted: **chemotherapy**, via bone marrow suppression. It recovers once chemo stops.

⚠ TRAP — **Stress alone can raise your WBC.** He said that when a count sits at the high end, he asks about the day of the draw — a stressful day can naturally double it. A high WBC is not automatically infection.

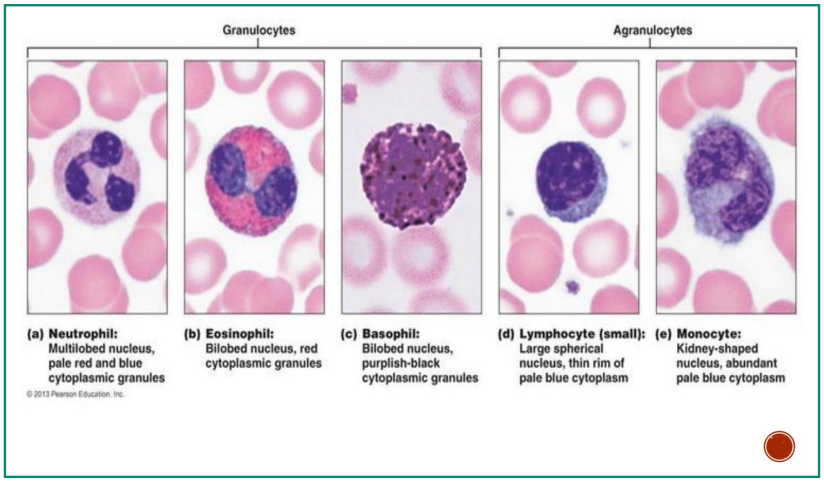
⚠ TRAP — **WBC count alone CAN indicate sepsis.** A student asked whether you must wait for culture. His answer: no — "definitely alone can be indicator." Sepsis kills in hours via blood pressure collapse, so clinicians start **broad-spectrum antibiotics immediately** and let culture refine it later.

Granulocyte vs agranulocyte

This split is a **historical artifact of light microscopy** — it records which granules were *visible* under the optics of the day, not which cells have granules.

Granulocytes (visible granules)	Agranulocytes (no visible granules)
Neutrophils · Eosinophils · Basophils	**Lymphocytes** (T, B) · **Monocytes** · NK cells

⚠ **TRAP — NK cells DO have granules** — they're classed as agranulocytes only because the granules aren't visible under light microscopy. He raised this unprompted. Also note he corrected a student who thought all five types were granulocytes: **lymphocytes and monocytes are agranulocytes.**



The nucleus is the identifier: neutrophil multi-lobed · eosinophil & basophil bi-lobed · lymphocyte large and round · monocyte kidney-shaped. Note WBCs are larger than the surrounding RBCs.

5 · Platelets

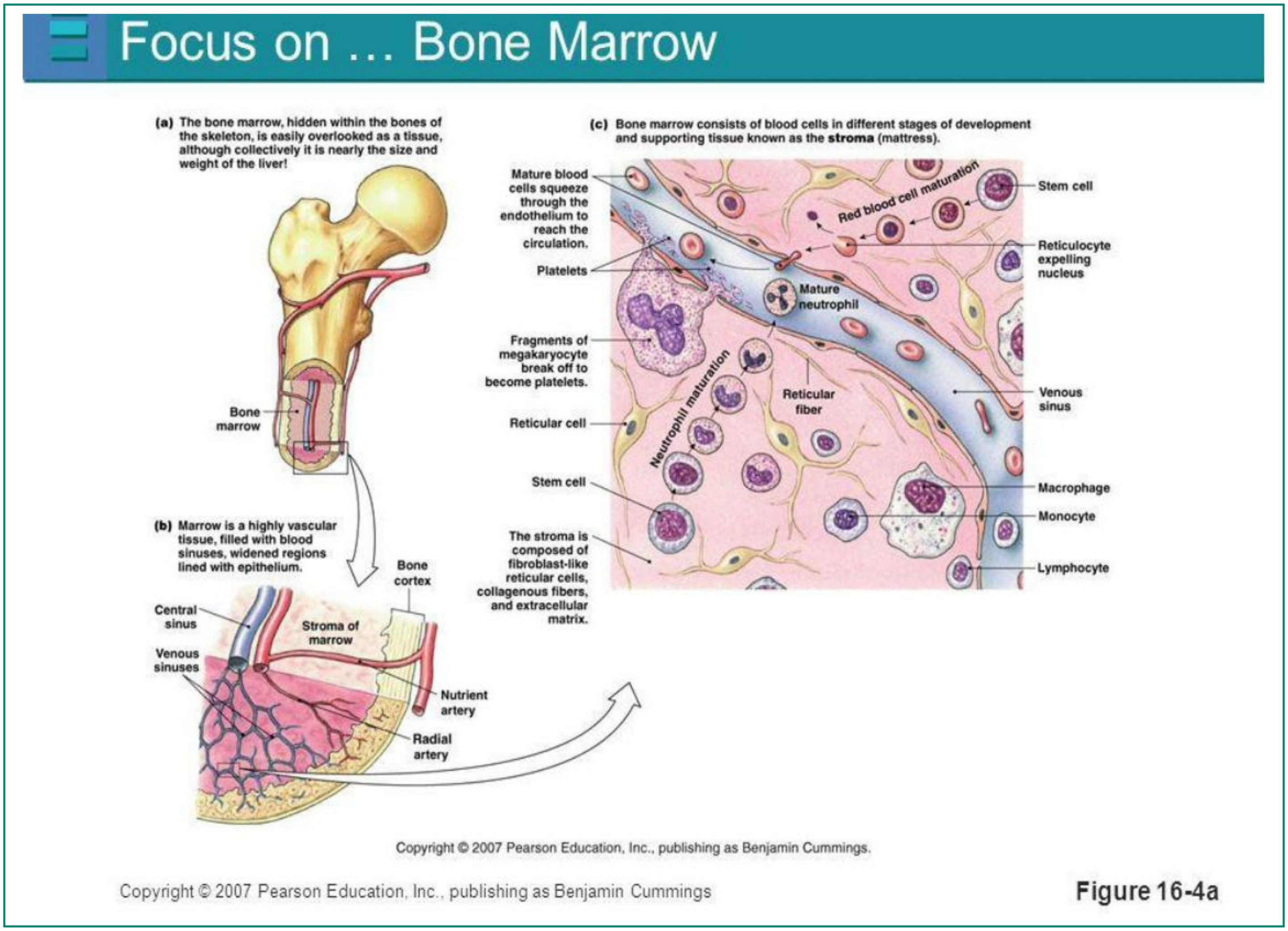
★ **HIGH YIELD — Platelets are not cells.** They are **cytoplasmic fragments** broken off a much larger marrow cell, the **megakaryocyte** (mega = large). No nucleus → they age fast → removed within ~10 days if unused.

	RBC	WBC	Platelet
A true cell?	No — no nucleus	**Yes** — nucleus + all organelles	No — a fragment
Lifespan	100–120 days	Hours to years	~10 days
Hormone	Erythro**poietin**	—	Thrombo**poietin**
Cleared by	**Spleen** (liver, less)	—	Spleen

In his words — He asked the class to name the RBC hormone from memory before giving the platelet one — because the naming is systematic: **erythro** (red) + poietin, **thrombo** (platelet) + poietin. Same for thrombocyte, thrombocytopenia, thrombocytosis.

Where platelets actually come from

"Cytoplasmic fragments of a megakaryocyte" is an abstraction until you see it. He zoomed into the femur, then into the marrow, then into the tissue itself — this is the last magnification.



Look for: the **megakaryocyte** pressing against the sinus with fragments breaking off — those fragments *are* the platelets. Also spot the **reticulocyte expelling its nucleus** (§3) happening in the same frame.

Marrow is highly vascular. Cells mature in the stroma, then squeeze through the endothelium into the venous sinus to reach circulation — which is how anything made here gets out.

Why platelets don't clot you to death

Platelets circulate through kilometres of vessel every day without sticking to anything. Something must be actively holding them off — and that something is **nitric oxide**, secreted continuously by the **endothelial cells lining the vessel wall**.

So activation isn't a switch that gets turned on. It's a suppression that gets **interrupted**: damage the wall, expose the collagen underneath, and the platelets respond.

> ⚠ **TRAP** — A student asked whether more nitric oxide is therefore better. His answer: **no — the normal amount**. “Nothing to say the higher is the better or the lower is the better. Just always the normal amount.” Beware any option phrased as “more X is better.”

6 · Hemostasis

hemo = blood, **stasis** = standing still. The system has to solve two opposite problems: clot **fast** when a vessel breaks, and clot **never** when it doesn't.

Acupuncture: press, never massage — steps 2–3 are fragile for the first minute. Massage tears the mesh → next-day bruise.

1 · VASCULAR SPASM
Vessel constricts
Least blood reaches the cut.
most important factor
★ he called this the

2 · PLATELET PLUG
~1 minute
Collagen exposed → platelets stick, swell, release ADP → recruit more.
but still fragile.
Loose plug. Blocks RBCs

3 · FIBRIN MESH
Coagulation cascade
Fibrinogen → fibrin threads trap RBCs + platelets.
and permanent.
Now it is sealed

Three steps, strictly in order. Step 1 does the most work; step 3 makes it permanent.

> ★ **HIGH YIELD — Vascular spasm is "probably the most important factor"** — his words — because it is immediate and cuts flow to the injury before anything else can happen. But it is never sufficient alone; a small leak persists, which is why steps 2 and 3 exist.

The vocabulary pairs that get tested

Inactive form	→	Active form	What changes
Fibrinogen	→	**Fibrin**	Soluble plasma protein → insoluble threads that mesh
Prothrombin	→	**Thrombin**	Inactive → the enzyme that converts fibrinogen

The **-ogen** ending signals the precursor. Fibrinogen was already sitting in the plasma list back in §2 — the coagulation cascade is what finally activates it.

Clinical — the acupuncture question he built the class around

He posed this as an open question and let three students answer before resolving it. The needle passes through dermis and hypodermis; a small vessel is very likely severed; a drop of blood appears on withdrawal. **Press, or press and massage?**

> ★ **HIGH YIELD — Press and hold. Never massage.** The platelet plug and fibrin mesh form within the first minute and are **still fragile**. Massage tears them → re-bleeding → and here's the detail he added: **you may see nothing at the time and a big bruise the next day.** Press ~30 seconds; once bleeding stops, you're done.

This is why hemostasis is on a WM360 syllabus at all — it's the mechanism directly under your hands every time you withdraw a needle.

7 · When Clotting Fails — Both Directions

Too much clotting

THROMBUS
Clot that forms and STAYS in an unbroken vessel.
No trauma. e.g. DVT — legs hanging, blood pools.

Manageable — clot-dissolving drugs, hospital.

breaks ▶ free

EMBOLUS
Clot now TRAVELING in the bloodstream.
Lodges in a vessel too small to pass.

LUNG	Pulmonary embolism
BRAIN	Stroke / CVA
HEART	Myocardial infarction

Why seniors are told to walk the aisle on long flights — the gastroc pump keeps blood moving, so no clot forms.

Same clot, two names. It becomes an embolus the moment it breaks free — often when the patient starts walking.

> ★ **HIGH YIELD — Thrombus vs embolus is a definition pair, and the difference is motion.** Thrombus = forms and **persists in an unbroken vessel** — no trauma. Embolus = the same clot once it **travels**. DVT is the classic thrombus.

The DVT chain: long flight, legs hanging, no gastrocnemius contraction → blood pools in the lower limbs → platelets aggregate with no injury at all → thrombus. Walking the aisle works because the calf muscle physically pumps venous blood back toward the heart. Crossed legs make it worse.

> **In his words** — The cruel irony he pointed out: a stationary DVT is **manageable** — hospital, clot-dissolving drugs, the patient survives. The danger comes the moment **the patient starts walking** and the clot breaks loose.

Too little clotting

Disorder	The broken link	Chain
Thrombocytopenia	Not enough platelets	Often **autoimmune** — the body destroys its own platelets

Disorder	The broken link	Chain
Hemophilia	A missing clotting factor	Genetic. **Factor VIII** (A) is more common; **factor IX** (B)
Liver disease	Factors never get made	★ see below — his trick question

> **★ HIGH YIELD — Why does liver failure cause bleeding?** He asked this deliberately, and a student guessed erythropoietin (wrong — that's kidney). The chain: **clotting factors are proteins → the liver makes proteins → severe liver disease → no fibrinogen or prothrombin → bleeding.** This connects straight back to §2. Expect it.

The three anticoagulants

Drug	Mechanism	Route / when
Aspirin (81 mg “baby”)	Blocks platelet **aggregation**	Oral, daily. Age 50+, family history of CV disease
Heparin	Inhibits **thrombin** → fibrinogen never becomes fibrin	**Injectable** — in hospital
Warfarin / Coumadin	Interferes with **vitamin K**	**Tablet** — continues after discharge

> ⚠ **TRAP — Aspirin does not destroy platelets**, and no blood thinner touches RBCs. He was explicit: “not destroying platelets, just the ability for aggregation kind of reduces.” The count stays normal; the stickiness drops.

Why vitamin K works as a target: clotting is a cascade — every link must be present. Vitamin K is one link, so removing it collapses the whole chain.

> **★ HIGH YIELD — Newborns get a vitamin K injection at birth.** Vitamin K is made by **gut bacteria**, and a newborn's gut hasn't been colonised yet. It's prophylaxis — so that any injury doesn't become fatal.

> ○ *Reference only (he said don't memorize): The full coagulation cascade and the thromboplastin→thrombin detail — “this is more for your review… we just limited to this much.” Know which drug hits which target; skip the intermediate steps.*

血行不已 · *“Blood flows without ceasing”*